

Public Rules, Executed Behavior, and Welfare in a Language-Model Economy

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Abstract

Public natural-language rules for language-model agents are usually judged by whether agents perform the action the rule names. We test that practice in the Verification Budget Economy, an eight-agent random-matching economy with scarce verification checks. A public announcement naming Easy-to-Hard gifts raises welfare by 2.06 points per account over a no-recommendation announcement (14/14 paired seeds). An exact accounting identity assigns the largest positive component to Hard–Hard check swaps—mutual cooperation in an unnamed stage game that is a Prisoner’s Dilemma—while the named relation contributes smaller positive terms and dominated Hard-to-Easy giving induced by the same text subtracts. Under the same gift announcement, blocking Hard–Hard transfers after the model responds lowers welfare by 3.21 (12/12 seeds). Replaying the frozen traces through the engine, we compare first decisions at states that are byte-identical except for the announcement: the named gift is engaged, and both the gift text and a harmful Easy–Easy directive (executed in 113 of 114 Easy–Easy meetings) raise Hard–Hard giving directly. In later meetings named giving drops sharply, while Hard–Hard giving stays high through the final round. Among the tested packages, explicit exclusion clauses reduced specific off-scope responses; packages that only named a relation did not confine responses to it.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; *Agent / discrete models*; *Natural language processing*.

Keywords

language-model agents, multi-agent systems, public rules, norms, causal mechanisms, welfare, agent-based simulation

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1 Introduction

Teams, markets, and simulations of language-model agents increasingly coordinate through public rules written in natural language. The usual check of such a rule is whether agents perform the action it names and whether an aggregate outcome improves. That check conflates four objects: the proposals agents make, the transactions

the environment executes from pairs of proposals, the state those transactions produce, and the welfare that results. A rule can be followed in its named relation and still owe its effect to a relation it never mentions.

We study this gap in the Verification Budget Economy (VBE), where Hard agents need verification checks, Easy agents have spare ones, and the engine executes only compatible pairs of proposals. Three prospectively frozen studies with a hosted model supply the causal comparisons; zero-call reanalyses of their frozen traces supply the accounting, the fixed-state comparisons, and descriptive closed-loop diagnostics. The results form one chain:

- *Most of the welfare gain is accounted for by an unnamed relation.* The engine implies that a Hard–Hard meeting is a strict Prisoner’s Dilemma and that giving in an Easy–Hard meeting is individually dominated for both roles. A gift announcement naming Easy-to-Hard gifts raises welfare. An exact accounting identity assigns its largest positive component to Hard–Hard swaps, smaller positive terms to the named relation, and a negative term to wasteful Hard-to-Easy gifts. Under the same announcement, blocking Hard–Hard transfers lowers welfare below the never-transfer benchmark (§4.1).
- *The text acts on unnamed roles directly.* Because each prompt is a function of reconstructable state, we can find first decisions taken from states that are byte-identical across arms except for the announcement. There, the named Easy-to-Hard gift is engaged, and Hard–Hard giving is raised by the gift text and by a harmful Easy–Easy directive. Among the tested packages, explicit exclusion clauses reduced specific off-scope responses; packages that only named a relation did not confine the response (§4.2).
- *Closed-loop behavior differs by relation.* Named giving drops sharply after an Easy agent’s first meeting, whereas Hard–Hard giving stays high through the final round, where giving is not a best response to the stage-game payoffs, and harmful Easy–Easy compliance does not decay. These patterns are descriptive; their mechanisms are not identified (§4.3).

The contribution is an identification design and the result it produces, not a reminder that intention differs from execution. Execution-layer intervention under a fixed announcement, exact transaction accounting, fixed-state comparison from frozen logs, and an engaged harmful control together show that the intuitive explanation—that the rule works because agents do what it says—accounts for little of the welfare effect here, and locate both the main welfare channel and the failure modes. The claim concerns a hosted model population in one economy, not private belief, common knowledge, or a self-enforcing institution.



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Table 1: Expected direct payoff change of one executed transfer relative to no transfer, derived from the engine. A sale moves the same payoffs as a gift in the same direction because marks are valueless.

Executed transfer	Giver	Receiver	Total
H–H swap	+1.83	+1.83	+3.66
E–H swap (E is “giver”)	−0.50	+1.83	+1.33
E–E swap	−0.50	−0.50	−1.00
One-way E→H	−0.50	+2.33	+1.83
One-way H→H	−0.96	+2.33	+1.37
One-way H→E	−0.96	0.00	−0.96
One-way E→E	−0.50	0.00	−0.50

2 Environment

2.1 Rules and state

VBE has $n = 8$ persistent accounts and $T = 24$ rounds; the supplement lists every parameter. All accounts start with score 0, and four random accounts start with one *mark* each. Marks have no direct or terminal value and persist across rounds. Each round proceeds as follows.

- (1) *Roles and endowments.* Exactly four accounts are drawn Hard and four Easy. Every account receives $B = 1$ check. An Easy account is paid $R = 3$ immediately.
- (2) *Matching.* Accounts are shuffled into four pairs, and each pair meets independently with probability $q = 0.4$, so an account has at most one meeting per round and roles are independent of pairing.
- (3) *Proposals.* Each party to a meeting simultaneously proposes `giveCheck`, `giveChits` (0 or 1 marks), and `requireChit`.
- (4) *Resolution.* In priority order: if both give and neither requires a mark, the checks are *swapped*; if one gives while requiring a mark and the other offers one, a *sale* executes; if one gives without requiring a mark, it is a *one-way gift*; otherwise nothing moves. A received check is a verification source; it is not added to the recipient’s inventory.
- (5) *Settlement.* A Hard account that received a check uses it and succeeds with probability $p_P = 0.93$; otherwise it uses its own check and succeeds with probability $p_H = 0.32$. Success pays R . Every check still in inventory pays $v = 0.5$, and checks expire.

Scores, marks, and each account’s last $K = 4$ meetings persist. The prompt shows the round, the agent’s role, checks, marks, and score, the partner’s role, checks, and marks, and the agent’s own last four meetings; it never shows the partner’s score or history. One payoff draw per account per round is pre-generated from the seed, so arms sharing a seed share roles, pairings, and payoff draws.

2.2 Transaction payoffs and stage games

Table 1 lists the expected change for every transfer the engine can execute. Each entry is checked against the engine by integrating its settlement over the payoff-draw strata, and an exhaustive test covers all 4,096 local meeting states (§3.1). Two rows matter most. An E–H *swap* is worth 1.33, not 1.83: the Hard agent gives away a

Table 2: Expected round payoffs (row, column) in a Hard–Hard meeting implied by Table 1. Keeping is strictly dominant and mutual giving maximizes welfare: a strict Prisoner’s Dilemma.

	Partner keeps	Partner gives
Keep	(0.96, 0.96)	(3.29, 0.00)
Give	(0.00, 3.29)	(2.79, 2.79)

check it would otherwise salvage, and the Easy agent cannot use the check it receives. A one-way H→E gift destroys 0.96.

The identities imply the stage games. An H–H meeting is a strict Prisoner’s Dilemma (Table 2): keeping is dominant, and mutual giving maximizes welfare. In an E–H meeting, keeping is strictly dominant for both roles, and the welfare-maximizing outcome is a one-way E→H gift, which costs the Easy agent 0.5. Every welfare-improving transfer therefore requires an agent to give up private payoff. The population game is repeated under random rematching with private four-meeting histories, where cooperation can in principle be sustained by community enforcement [14, 25]; our design does not identify such a mechanism.

Two model-free benchmarks run on the same seeds, schedules, and draws. The *ex-ante first best* maximizes expected welfare meeting by meeting over the engine’s action space without using realized draws: an H–H swap, a one-way E→H gift, and no transfer otherwise. It is a planner benchmark, not an equilibrium. The *ex-post first best* uses realized draws and bounds attainable welfare. *Never-transfer* sets the baseline.

3 Studies and Inference

All model studies use `deepseek-flash` in one provider-labeled serving era at temperature zero. Every protocol, seed list, prompt, contrast, and gate was hash-frozen locally before its first target call; none was externally registered. Each study is bracketed by a 14-call provider-health check before the first and after the last target call, with external controls, dominance anchors, entropy, returned-model, catalog, and fingerprint gates; all brackets reported here are healthy and identity-consistent. The supplement records the anomalous earlier service window that motivated the bracket.

W-CO (discovery). Gift versus money announcements on 18 fresh paired seeds. Its welfare contrast was prospective; its planned gift-for-sale substitution gate failed, and the H–H channel was found afterwards. We use it only as discovery and as a third replication of the fixed-state results.

W-RG (execution intervention). Four arms on 12 fresh paired seeds: neutral; gift; gift with H–H check giving blocked; gift with unconditional E→H give proposals blocked. The three gift arms show the model the same announcement. After the model responds, a filter sets only `giveCheck` to false on eligible proposals and leaves the other fields unchanged; the engine then executes as usual and later prompts reflect the changed history. W-RG estimates the dynamic total effect of changing the execution contract under a fixed announcement.

W-SGB (package boundary). Seven announcements on 14 fresh paired seeds under unmodified execution (Table 3). The harmful E–E directive names a transfer that destroys 1.0 per execution and carries its own frozen engagement gate. Packages other than the three historical anchors are not length-matched, so contrasts identify whole packages.

Inference. Let $Y_s(d, e)$ be the seed- s population outcome under announcement d and execution contract e ; the seed fixes initial state, roles, pairings, and payoff draws. W-RG and W-SGB estimate, respectively,

$$\tau_{RG} = \frac{1}{12} \sum_s [Y_s(d_{\text{gift}}, e_{\text{std}}) - Y_s(d_{\text{gift}}, e_{\text{HH}})],$$

$$\tau_{SGB} = \frac{1}{14} \sum_s [Y_s(d_1, e_{\text{std}}) - Y_s(d_0, e_{\text{std}})].$$

Both include downstream feedback through holdings and histories, so the inferential unit is the paired seed-level trajectory, not a meeting or account, and neither identifies a natural indirect effect [6, 23]. We report percentile bootstrap intervals (20,000 resamples of seed differences; pointwise, not simultaneous) and one-sided exact sign-flip p -values for $\Delta > 0$ under a zero-centered sign-symmetry null. W-SGB applies Holm correction at 0.025 within its eight H–H and eight welfare contrasts. A frozen gate passes when the mean reaches the minimum relevant effect size (MRES: 0.25 for swap rate, 0.5 for welfare) and the adjusted $p \leq 0.025$. Passing a gate shows a large estimate and a nonzero effect; it does not show $\Delta > MRES$. We therefore add supplementary exact tests of $H_0: \Delta \leq MRES$ and leave-one-seed-out ranges. With 12 or 14 seeds the smallest attainable one-sided p is 2^{-12} or 2^{-14} , reached whenever every seed agrees in sign, so we report sign counts throughout. Decision-level rates are pooled over decisions, and their intervals resample whole seeds (stratified by study) and recompute the pooled rate; no decision or meeting is treated as an independent unit.

3.1 Zero-call reanalysis of frozen traces

The reanalysis adds no model call and changes no frozen artifact. We replay every frozen run through the frozen engine by feeding back its recorded proposals. Because roles, pairings, and draws come from the seed alone, the replay reproduces all 182 runs exactly (every final score and every meeting resolution), recovering the exact prompt behind each of the 14,010 decisions, a count equal to the three studies’ frozen call totals. This supports three analyses. *Accounting:* each run’s welfare equals its never-transfer welfare plus the realized contribution of each executed transfer, computed from the regenerated payoff draws; the identity holds exactly in all 182 runs. *Fixed states:* at an agent’s first meeting, its prompt carries no treatment-dependent history, and the reconstructed prompts are byte-identical across arms once the announcement is removed (verified for every first meeting); comparing arms there isolates the announcement’s effect on that decision. *Engine validation:* an exhaustive test enumerates every role pair, inventory, and proposal combination (4,096 cases) and checks resource conservation, the filters, and settlement against Table 1. *Closed-loop diagnostics:* a second pass tabulates giving by exact round, by meeting index, and by the agent’s own previous outcome, with seed-clustered intervals. These analyses are post hoc and descriptive with respect to the frozen studies. The replay verifies the user prompt. The rest of each request (a fixed system prompt stating the rules, temperature 0, a

64-token limit, JSON mode, reasoning disabled, no tools) is set by constants in the hash-frozen execution code; request bodies and their hashes were not retained, so identical provider requests are implied by that code rather than verified from logs.

4 Results

4.1 Most of the welfare gain is accounted for by an unnamed channel

The gift announcement raises welfare. In W-SGB it exceeds neutral by 2.0625 points per account ([1.429, 2.813], 14/14 seeds), raising the H–H swap rate by 0.607; W-CO’s gift-minus-money contrast (+2.333, 17/18 seeds) and W-RG’s gift-minus-neutral contrast (+1.682, 9/12 seeds) agree in sign (Table 5).

Where the gain comes from. The named one-way transfer is rare in the realized trace. In W-SGB’s gift arm, 105 of 125 H–H meetings end in swaps (27 under neutral), while only 9 of 304 E–H meetings end in a one-way E→H gift. Figure 1B decomposes the welfare difference exactly. H–H swaps contribute +2.20 per account ([1.53, 2.92]), the largest positive component. The named relation adds smaller positive terms: E–H swaps +0.36 ([0.17, 0.57]) and one-way E→H gifts +0.08 ([0.00, 0.16]). Ninety-nine one-way H→E gifts (3 under neutral) cost 0.62 ([0.46, 0.80]). W-RG gives the same pattern (+2.41 from H–H swaps, −1.13 from H→E gifts). The dominant positive component of the gain is therefore cooperation in the unnamed Prisoner’s Dilemma, not the unilateral gift the announcement names, and the same text induces a dominated, welfare-destroying transfer that offsets part of it. Reciprocity also costs welfare in the named relation: in W-RG’s gift arm, 25 of 26 feasible E→H gift intentions met an unconditional Hard give and resolved as E–H swaps, each worth 0.5 less than the one-way gift (12.5 points in total). Joint proposals show where these transfers come from: in the gift arm’s E–H meetings, only the Easy agent gave in 9, only the Hard agent in 99, both in 28, and neither in 168. Table 4 places every package against model-free benchmarks on the same seeds. The gift package captures 35% of the ex-ante feasible gain; the Hard-partner package captures the most, 38%, because it keeps H–H swaps (115) while nearly eliminating H→E gifts (18). Mutual H–H giving is not simply two independent proposals: under neutral it occurs in 27 of 125 meetings, against about 8 expected from the marginal give rate, whereas under gift it matches independence (105 versus 104).

Execution intervention. W-RG holds the announcement fixed and removes the H–H channel after the model responds. Welfare falls by 3.208 ([2.469, 3.891], 12/12 seeds, Figure 1A), from 55.30 to 52.09—below the same seeds’ never-transfer benchmark (52.53) in 10 of 12 seeds. Without H–H circulation, the remaining transfers induced by the gift announcement are net negative. Removing transfers worth 3.66 each will lower welfare; the result is informative because the removed channel is the one the announcement does not name, and because what remains of the announcement’s effect is wasteful. The E→H filter changed little: of its 20 blocked proposals, 16 would have produced E–H swaps and instead executed as one-way H→E gifts, and 4 would have produced one-way E→H gifts. The regime

Table 3: W-SGB announcement packages. Every package also states that it is a public statement rather than an engine rule, may be followed or ignored, and that agents should maximize their own score. The supplement reproduces every package verbatim.

Arm	Distinguishing clause (abridged)	Named relation; explicit exclusion
Neutral	“announces no exchange recommendation”	none
Gift exact	“an Easy agent gives a leftover check to a Hard agent”	$E \rightarrow H$ gift
Easy only	“only an Easy agent gives ... A Hard agent does not give”	$E \rightarrow H$ gift; excludes Hard actors
Any holder	“any agent who holds a leftover check gives it to a Hard agent”	$\{E,H\} \rightarrow H$ gift
Hard partner	“an agent gives ... only when the current partner is Hard ... do not give ... when the current partner is Easy”	$\{E,H\} \rightarrow H$ gift; excludes Easy partners
Money exact	“a mark is accepted as payment ... a Hard agent who holds a mark may buy”	$E \rightarrow H$ sale
Harmful E-E	“when two Easy agents meet ... each gives that check to the other”	$E \leftrightarrow E$ swap

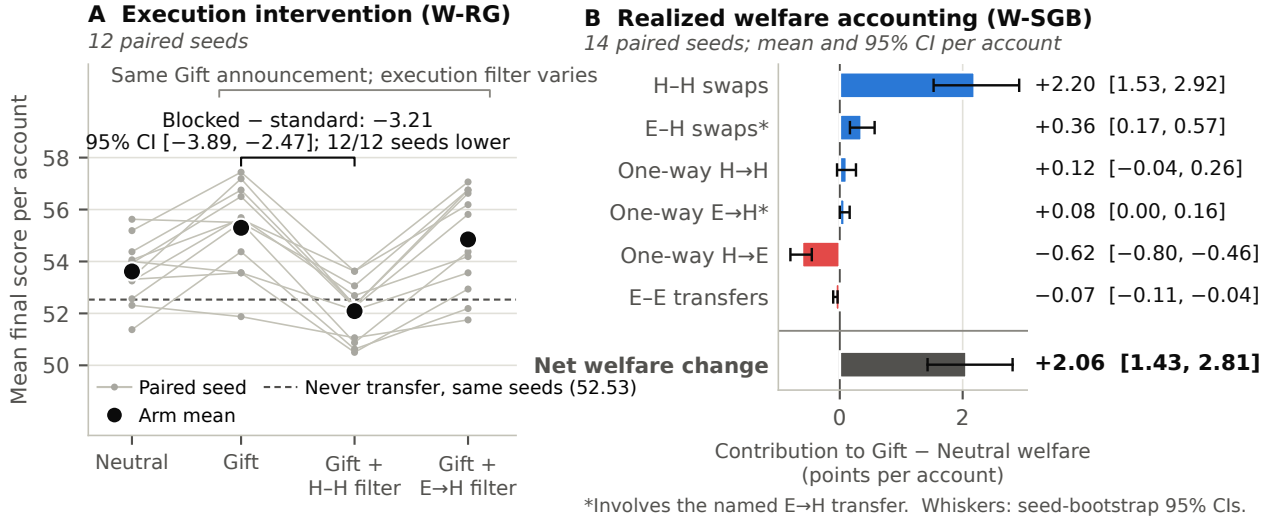


Figure 1: (A) Mean final score per account in W-RG under four execution regimes. Thin lines connect each seed’s four paired runs; large markers are arm means; the dashed line is the same seeds’ never-transfer benchmark (the ex-ante first best, 61.81, lies above the plotted range). The bracket gives the paired difference between the H–H-filtered and standard gift arms with its seed-bootstrap 95% CI. Neutral is an external reference; the three gift arms share the announcement, not the full prompt, because later prompts carry realized history. Blocking H–H transfers lowers welfare within the gift announcement condition; W-RG has no neutral arm with H–H blocked. **(B) Realized welfare accounting of the W-SGB gift-minus-neutral difference by executed transfer type, computed from actual transfers and payoff draws.** Bars are seed means per account; whiskers are seed-bootstrap 95% CIs, and the net interval is the frozen contrast interval (Table 5). H–H swaps are the largest positive component; rows marked * involve the named E→H transfer. This is an accounting identity, not a mediation estimate. Rounded components may not sum to the displayed total. The W-RG decomposition is in the supplement.

contrast “E→H-blocked minus H–H-blocked” (+2.760, 12/12) therefore compares keeping H–H circulation with keeping the little that the E→H path carried.

W-RG has no neutral-with-H–H-blocked arm. Writing $Y_{d,e}$ for mean welfare under announcement $d \in \{G, N\}$ and execution $e \in \{S, B\}$, the data fix $Y_{G,S} - Y_{N,S} = 1.68$ and $Y_{G,S} - Y_{G,B} = 3.21$, hence $Y_{G,B} - Y_{N,S} = -1.53$, but not $\tau_B = Y_{G,B} - Y_{N,B}$. Never-transfer is a scripted benchmark, not neutral-blocked; equating the two would give $\tau_B \approx -0.44$, which we use only as a prior for sizing a new study. W-RG therefore shows that blocking H–H transfers lowers welfare within the gift condition; it does not identify how the gift’s gain over neutral depends on the H–H channel. The accounting identity is consistent with strong dependence but is

not a causal decomposition. The supplement specifies the four-arm factorial on fresh seeds that estimates τ_S , τ_B , and $I = \tau_S - \tau_B$.

The stricter supplementary test confirms every passing frozen gate, with one exception: the harmful directive’s H–H effect rejects $\Delta \leq 0.25$ before correction ($p = 0.0089$) but not after Holm (0.027). Every passing contrast keeps its mean above MRES when any single seed is dropped.

4.2 The text acts on unnamed roles directly

Aggregate role-conditional proposal rates mix two sources: the announcement changes what an agent does in a given state, and it changes which states agents reach. First decisions remove the second source. Figure 2A shows, for every W-SGB package, the

Table 4: Absolute welfare and executed transfers by W-SGB package (14 seeds, same schedules and payoff draws). Gain is welfare minus the never-transfer benchmark; share is gain divided by the ex-ante first best’s gain. Transfer columns count executed meetings out of 125 H–H, 304 E–H, and 114 E–E meetings per arm.

Package or benchmark	Welfare	Gain	Share	H–H swaps	E–H swaps	One-way E→H	One-way H→E	E–E swaps
Neutral	55.28	1.04	0.12	27	0	0	3	0
Gift exact	57.34	3.10	0.35	105	28	9	99	2
Easy only	55.23	0.99	0.11	29	0	0	16	1
Any holder	57.25	3.00	0.34	111	28	7	95	0
Hard partner	57.68	3.44	0.38	115	1	4	18	0
Money exact	55.07	0.83	0.09	21	0	0	0	0
Harmful E–E	55.46	1.22	0.14	75	1	3	35	113
Never transfer	54.24	0	0	0	0	0	0	0
Always give	60.81	6.57	0.73	125	304	0	0	114
Ex-ante first best	63.19	8.95	1	125	0	304	0	0
Ex-post first best (uses realized draws)	63.26	9.02	1.01	—	—	—	—	—

Table 5: Prospective paired contrasts. CIs are pointwise percentile bootstrap intervals over seeds; +/–/0 counts seeds by sign. p is the one-sided exact sign-flip p for $\Delta > 0$, Holm-adjusted within each W-SGB endpoint family. p_{MRES} tests $H_0: \Delta \leq MRES$ (supplementary; Holm within each W-SGB family). Frozen gates require mean $\geq MRES$ and $p \leq 0.025$.

W-RG (12 seeds)	Δ	95% CI	+/–/0	p	p_{MRES}	Gate
Gift – neutral, welfare	1.682	[0.854, 2.463]	9/3/0	.0034	.011	pass
Gift – H–H-blocked, welfare	3.208	[2.469, 3.891]	12/0/0	.00024	.00024	pass
E→H-blocked – H–H-blocked, welfare	2.760	[2.094, 3.412]	12/0/0	.00024	.00024	pass
Gift – H–H-blocked, H–H swap rate	0.753	[0.633, 0.862]	12/0/0	.00024	.0020	pass

W-SGB (14 seeds)	H–H swap rate (MRES 0.25)					Welfare (MRES 0.5)				
	Δ	95% CI	+/–/0	p	p_{MRES}	Δ	95% CI	+/–/0	p	p_{MRES}
Gift – neutral	.607	[.510, .714]	14/0/0	.00049	.00049	2.063	[1.429, 2.813]	14/0/0	.00049	.00085
Gift – money	.663	[.568, .755]	14/0/0	.00049	.00049	2.268	[1.670, 2.960]	14/0/0	.00049	.00049
Gift – harmful E–E	.229 [†]	[.105, .351]	10/3/1	.0054	1.0	1.875	[1.348, 2.433]	14/0/0	.00049	.0011
Harmful E–E – neutral	.378	[.287, .474]	14/0/0	.00049	.027	.188	[–.339, .710]	9/5/0	.51	1.0
Gift – Easy only	.592	[.482, .697]	14/0/0	.00049	.00049	2.112	[1.464, 2.813]	14/0/0	.00049	.0011
Any holder – neutral	.658	[.549, .771]	14/0/0	.00049	.00049	1.969	[1.295, 2.696]	14/0/0	.00049	.0011
Hard partner – neutral	.703	[.596, .810]	14/0/0	.00049	.00049	2.406	[1.759, 3.107]	14/0/0	.00049	.0011
Money – neutral	–.056	[–.126, –.001]	1/5/8	.97	1.0	–.205	[–.339, –.071]	1/9/4	.996	1.0

[†]Significant but below the 0.25 MRES, so its frozen gate fails. Every other row with mean $\geq MRES$ and $p \leq 0.025$ passes its frozen gate.

unconditional-give rate at each agent’s first meeting, where the prompt differs across arms only in the announcement (18 H→H, 35 H→E, 39 E→H, and 20 E→E decisions per arm).

The named relation is engaged. Under gift exact, 17 of 39 Easy agents give to a Hard partner at their first meeting and none do under neutral (17 discordant pairs in one direction, 0 in the other). The same holds in W-CO (26 of 41, gift versus money) and W-RG (16 of 31). The low closed-loop rate of named giving (0.123 above neutral in W-SGB) therefore does not reflect a failure to engage the named relation; §4.3 shows that named giving drops sharply after the first meeting. A post-hoc engagement screen that compared closed-loop named rates with 0.25 would call the gift package unengaged and the any-holder and Hard-partner packages engaged; at first decisions the classification reverses, and in closed loop it flips again between thresholds of 0.10 and 0.15 (supplement). We therefore treat that screen as descriptive only.

Cross-role giving is a direct text effect. At first decisions, Hard agents meeting a Hard partner give at rate 0.39 under neutral, 0.94 under the gift package, and 0.83 under the harmful E–E directive, which names only Easy agents. All disagreements with neutral run

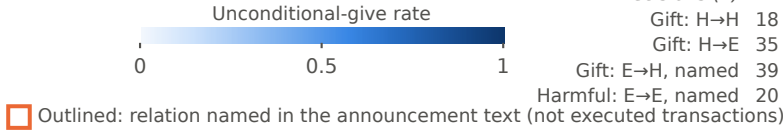
one way (10/0 and 8/0 discordant decisions; seed-clustered 95% intervals of the net discordance rate [0.27, 0.83] and [0.22, 0.65], from the 8 seeds with such decisions). The harmful directive’s H–H spillover is therefore not a by-product of the trajectories it creates. Under the gift package, Hard agents also give to Easy partners (0.69), outside both the named actor and the named beneficiary.

Exclusion clauses reduced specific off-scope responses. Among the tested packages, those with explicit exclusions reduced the excluded response at first decisions: “do not give when the partner is Easy” reduces H→E giving to 0.00 (from 0.69 under gift exact), and “a Hard agent does not give” leaves H→H giving near the neutral level (0.44 versus 0.39). Packages that only *name* a relation did not confine the response to it: any holder names Hard beneficiaries, yet Hard agents give to Easy partners at rate 0.63. These are reductions of announcement-induced responses, not strict scope enforcement: over full runs the Hard-partner package still executes 18 one-way H→E gifts, and the Easy-only package retains 29 H–H swaps, close to neutral’s 27 (Table 4). Restriction can also remove intended giving: the Easy-only package suppresses its own named relation (0/39), so it cannot isolate actor exclusivity. Easy agents give at first decisions when the package names Easy actors (0.44),

A Matched first-decision states (W-SGB)

Columns: decider → partner. Cells: rate, gives/decisions

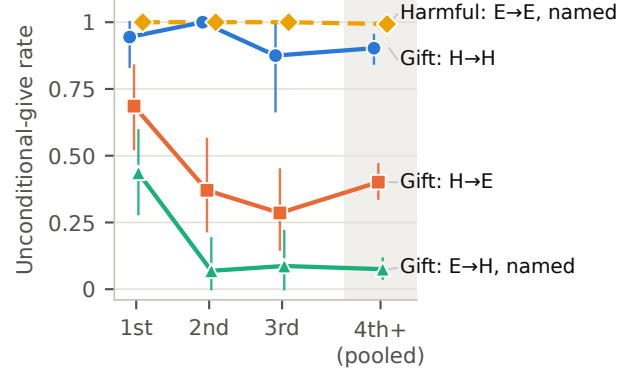
	H→H	H→E	E→H	E→E
Neutral	0.39 7/18	0.00 0/35	0.00 0/39	0.00 0/20
Gift exact	0.94 17/18	0.69 24/35	0.44 17/39	0.00 0/20
Easy only	0.44 8/18	0.06 2/35	0.00 0/39	0.00 0/20
Any holder	1.00 18/18	0.63 22/35	0.08 3/39	0.05 1/20
Hard partner	1.00 18/18	0.00 0/35	0.00 0/39	0.00 0/20
Money exact*	0.06 1/18	0.00 0/35	0.00 0/39	0.00 0/20
Harmful E-E	0.83 15/18	0.00 0/35	0.00 0/39	1.00 20/20



*Money exact names a sale, which this unconditional-give metric does not measure.

B Giving across successive meetings (W-SGB)

Observed trajectories; history is not randomized



Agent's meeting index in the run (not calendar round)

Decisions (n)				
Gift: H→H	18	23	24	185
Gift: H→E	35	27	35	207
Gift: E→H, named	39	29	23	213
Harmful: E→E, named	20	33	30	145

Figure 2: (A) Unconditional-give rate at each agent's first meeting in W-SGB (cells: rate and gives/decisions; first decisions from 14 paired seeds). Within each comparison with neutral, prompts are byte-identical except for the announcement. Cells measure one-sided proposals, not executed transactions; orange outlines mark the relation a package's text names, not compliance or welfare alignment. Money exact names a sale, which this metric does not measure. (B) Unconditional-give rate by the decider's meeting index in the run (not calendar round; 4th+ pools later meetings), W-SGB. Points pool all decisions in a bin, so first points equal panel A; whiskers are 95% intervals from resampling whole seeds; denominators are listed below the axis. Later meetings condition on realized history, so this panel is descriptive and does not identify a memory or learning effect. Neutral curves and other cells are in the supplement.

not when it names “any agent” (0.08) or “an agent” (0.00). Because the packages are not minimal textual contrasts, these comparisons identify package effects, not a linguistic rule. The money package lowers first-decision H→H giving (0.06, six discordant decisions all favoring neutral).

Noise floor. Temperature zero does not make the endpoint deterministic [7]. Across W-RG's three gift arms, decisions with byte-identical full prompts returned identical proposals in 87–90% of 877 comparisons, and disagreements ran in both directions (10 versus 16, 16 versus 15, and 13 versus 11); the seed-clustered interval of the net give-direction rate includes zero for every pair. Cross-announcement discrepancies at first decisions are instead predominantly one-directional (17/0, 10/0, 8/0), with seed-clustered intervals that exclude zero. The two sets differ in state composition, and the 877 comparisons share prompts across arm pairs, so we read this contrast descriptively. W-CO's first-decision comparisons are gift versus money, not gift versus neutral, and are not pooled with the others.

4.3 Closed-loop behavior

Figure 2B follows the same cells over each agent's successive meetings. The index counts an account's meetings in any role, and roles are redrawn each round, so a point does not track a fixed set of agents in one role. Everything in this subsection conditions on realized history, round, and partner, none of which is randomized; it is descriptive.

Named giving drops sharply after the first meeting. E→H giving falls from the first to the second meeting in all three studies (W-CO 26/41 to 1/45; W-RG 16/31 to 0/24; W-SGB 17/39 to 2/29) and stays low over all later meetings (23/332, 10/231, 20/265). The drop does not require earlier experience of the named relation: an Easy agent's first E→H decision gives at 0.06 (13/222 pooled over the three gift arms; seed-clustered CI [0.03, 0.09]) when it follows a meeting in another role, against 0.53 (59/111) when it is the account's first meeting. Pooling first meetings over rounds 1–2 against prior-meeting decisions (0.51 versus 0.05) confounds history with round, because round 1 cannot carry history. Within round 2 alone the rates are 0.45 (13/29) and 0.05 (1/20), and a round-standardized difference over rounds 2–6 is 0.44 ([0.28, 0.61]). This narrows the round confound but does not isolate the memory field: an earlier

meeting also changes the score, the agent’s own past action, and the prompt’s length. Named gifts still occur late (4 of 31 final-round E→H decisions).

H→H giving stays high in the final round. Under gift (standard execution, three studies), Hard agents gave to Hard partners in 27 of 30 final-round decisions (15 meetings in 15 seeds; both gave in 13), against 0.91 in rounds 1–21; the final-minus-earlier difference is -0.02 ($[-0.18, 0.10]$). The prompt states the round and horizon, and by Table 2 keeping gains 0.50 against a giving partner and 0.96 against a keeping one. High final-round giving with no apparent decline is therefore inconsistent with explanations that rely entirely on future incentives and final-round best responses. It does not identify what sustains giving: the data cannot separate misuse of the horizon, misrepresented payoffs, persistence of earlier actions, or an announcement-induced objective, and the final-round sample is too small to claim a zero endgame effect.

Giving after an unreciprocated transfer. Under gift, Hard agents whose previous meeting was an unreciprocated transfer give to their next Hard partner at 0.89 (94/106), but most of those losses occurred in H→E meetings. After an unreciprocated H→H transfer to a different partner the rate is 0.64 (9/14), and when the agent’s last H→H meeting was unreciprocated it is 0.73 (22/30), against 0.97 (365/377) after a mutual swap. Under the harmful directive, H→H giving after an unreciprocated transfer is 0.37 (7/19), against 0.83 at first meetings. Agents are rematched at random, so these rates mostly concern new partners and do not test retaliation against the account that failed to reciprocate (12 same-partner cases under gift). The data support only that, under some announcements, giving toward later Hard partners stays high after unrewarded transfers.

Harmful compliance persists. E→E giving under the harmful directive remains 0.99 at agents’ fourth and later meetings and 6/6 in the final round, although every execution costs each Easy agent 0.5 of salvage. High giving persists without any added exogenous sanction or enforcement; whether it relies on history-dependent responses is not identified. The supplement gives the exact-round, encounter-index, and post-loss tables with numerators, denominators, and seed counts.

4.4 A harmful directive and an unexecuted money package

The harmful directive adds no rule, reward, monitoring, or sanction and says it may be ignored, yet it executes 113 E→E swaps in 114 opportunities (frozen engagement contrast 0.982, $[0.946, 1.000]$, adjusted $p = 0.00006$). Its net welfare effect is $+0.19$ ($[-0.34, 0.71]$), because H→H swaps it induces ($+1.23$ per account) offset the 1.01 it destroys through named E→E swaps. A single public sentence thus induced persistent, directly costly resource transfers, but its net welfare effect is not distinguishable from zero, so the data do not show a net population harm. Behavioral spread across roles and welfare are separate quantities. Whether agents accept the salvage loss knowingly or fail to represent it remains open; a payoff-table test (supplement) can show whether payoff representation shapes the response, though not what agents privately understand.

No arm of any study executed a mark sale. The funnel locates the failure. Under money exact, the Hard partner held a mark in 140 of

304 E→H meetings, the Easy seller proposed a mark-contingent sale in all 140 (136 of 140 under neutral), and the Hard buyer offered a mark in none. Across all 13 arms of the three studies, Hard agents offered a mark in 0 of 3,443 decisions taken while holding one—including under a money announcement that tells mark holders they may buy. The funnel locates where the sale fails—buyer non-payment—but not why, so the money package tests nothing about monetary acceptance or priced semantics, and we make no claim about either.

5 Discussion

What is identified. Within the gift announcement, blocking the H→H execution channel lowers welfare (W-RG); how the gift’s gain over neutral depends on that channel awaits the factorial. The accounting identity assigns the largest positive component of the gift-versus-neutral gain to H→H swaps, smaller positive terms to the named relation, and a negative term to H→E waste. At fixed first-decision states the announcement directly raises giving by unnamed actors toward unnamed beneficiaries; among the tested packages, explicit exclusions reduced specific off-scope responses. In closed loop, named giving drops sharply after the first meeting while H→H giving and harmful compliance stay high through the final round; these are descriptive patterns whose mechanisms remain open. The best description of the object is a *distributed policy program*: public text changes a population’s mapping from role and state to proposals, and the engine determines which proposals become transitions. It is not an institution in the stronger sense of common belief, legitimacy, consent, or self-enforcement [35, 43], and every agent receives the announcement directly, so cross-role responses are population responses, not peer-to-peer contagion.

Design implications. Version natural-language rules like code and log the proposal, execution contract, and transition behind every outcome. Evaluate a rule on the relations it does not name, not only on the one it names. State exclusions explicitly and test them: here, packages with exclusion clauses reduced specific off-scope responses where positive scope predicates did not, yet neither enforced scope strictly, and one exclusion also suppressed the intended gift. Measure welfare against model-free benchmarks on the same schedules, because named compliance, cross-role spread, and welfare moved independently. Behavioral adoption is not a reliable proxy for welfare quality: the most compliant package destroyed value in its named relation, and the most effective package was not the one that named the welfare-maximizing transfer. Finally, when prompts are a function of logged state, frozen traces support exact replay, fixed-state comparison, and exact accounting at no inference cost; logging request hashes would also let them verify the provider requests.

6 Limitations

All model evidence comes from one hosted model in one provider-labeled serving era; health brackets bound the service interval but do not establish immutable weights or cross-model generality. The only cross-family evidence is three historical grok-4.5 seeds, too few to count as replication. Packages are not length- or token-matched. Fixed-state comparisons cover first decisions only; the endgame, exact-round, and post-loss analyses are descriptive and

do not manipulate horizon, history, or partner. Request bodies were not logged, so request-level identity rests on the frozen code. W-RG lacks the neutral-with-H-H-blocked arm needed to identify how the gift effect depends on H-H circulation. The economy fixes price, supply, horizon, and technology, and mean score is its only welfare criterion. The H-H endpoint was selected after W-CO’s planned mechanism failed; later studies test that selected phenotype on fresh seeds. The reanalyses in §3.1 are post hoc. The supplement specifies the next experiments in priority order: the four-arm announcement-by-execution factorial; a modular fixed-state study of announcement scope, memory content, remaining horizon, and payoff representation; a bounded replication of its core contrasts on a second model family, which could show only that those fixed-state effects replicate; and an optional paired test of the payoff to unilateral free riding.

7 Related Work

Norms and institutions. Institutional grammar and normative multi-agent systems formalize role-indexed obligations, permissions, monitoring, and sanctions [10, 13, 16, 21, 24, 26]; social-law synthesis and norm-emergence work study top-down constraints and bottom-up convergence [33, 41, 42, 45, 46]. These traditions specify what a rule means or authorizes. We ask what a natural-language rule executes when model policies interpret it, and find that its realized actor and beneficiary scope can exceed the written one; in the packages we tested, explicit exclusions reduced but did not eliminate that excess. Convention research separates shared observation from common knowledge [11, 12, 29, 40]; our announcement is common exposure, and we infer no epistemic state.

Language-model populations. Generative agent-based models, simulated samples, and economic agents [2, 4, 5, 15, 22, 30, 31, 36, 50] and studies of coordination, cooperation, conventions, and legal or economic structure in LLM collectives [1, 18, 32, 38, 39, 49, 51] show what such populations can do. Concordia separates agents’ described actions from outcomes a game master resolves [48]; GovSim pairs aggregate outcomes with communication ablations and capability probes [37]; Welfare Diplomacy scores general-sum welfare [34]. Our design adds three elements we have not found combined in this literature: execution-layer intervention under a fixed announcement, an exact transaction-level welfare identity, and comparison of decisions at states that are identical across treatments. Together they show that the named relation is not the main welfare channel. The naming-game debate shows that collective behavior can reproduce training data [8, 19], so we claim no emergence. Search-theoretic money motivates the token environment [27, 28]; fixed supply, horizon, and prices preclude a monetary-equilibrium claim.

Games with language models and human benchmarks. Repeated-game studies of LLMs usually state the payoff structure of a two-player game directly. Akata et al. play finitely repeated 2×2 games with simplified payoff descriptions and neutral action labels, and study responses to defection and prompt interventions [3]; Fontana et al. analyze payoff and history comprehension, memory windows, forgiveness, and retaliation in the iterated Prisoner’s Dilemma and

build behavioral profiles from logs [20]; CoopEval compares repetition, reputation, mediation, and contracts as mechanisms that sustain cooperation under individual payoff pressure [47]. In VBE, role-dependent local incentives are derived from resource-settlement rules rather than stated as a matrix, and an announcement aimed at one role relation is evaluated on unnamed relations and on population welfare. Human experiments suggest which dimensions to test next: remaining horizon [44], conditional cooperation [17], and the difference between empirical and normative expectations [9]. They are not a baseline for this environment, and a history effect in our logs could reflect reciprocity, imitation, or expected returns as well as norm compliance.

8 Conclusion

A public sentence changed a language-model population’s welfare, mostly not through the transaction it named. The largest positive component came from cooperation in an unnamed Prisoner’s Dilemma, net of waste the same sentence induced; the text acted on unnamed roles directly; and among the tested packages, explicit exclusions reduced specific off-scope responses without enforcing scope. Evaluating such rules requires auditing named compliance, cross-role response, executed transitions, and welfare as separate quantities.

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